



PROG1246D Alternative Web Technologies

Alternative Web Technologies Assignment 1

Employee Weekly Compensation Calculator TypeScript

1. Specifications

- Create TypeScript that contains the specified interface, parent and child classes as per the class diagram
- The Employee interface specifies the following contract:
 - i. `displayInformation`
 - 1. The implementing classes will implement a method that will display the child employee details
 - ii. `calculateCompensation`
 - 1. The implementing classes will implement this method using business rules specified below
 - iii. `saveEmployee`
 - 1. The implementing classes will implement this method and execute the data validation methods.
- The Employee class will validate:
 - i. The age of the employee must be greater than or equal to 16
 - 1. Display a console message upon failure
 - ii. The rank must be between 1 and five inclusive
 - 1. Display a console message upon failure
 - iii. The SSN must match the pattern: ###-###-#### (Use RegEx)
 - 1. Display a console message upon failure
- The Fulltime Employee Compensation is calculated with the following business rules within the `calculateSalary` method:
 - i. Overtime hours between 1 – 10 are paid at 1.25 times the hourly rate
 - ii. Overtime hours between 11 – 20 are paid at 1.5 times the hourly rate
 - iii. Overtime hours between 21 – 30 are paid at 1.75 times the hourly rate
 - iv. Overtime hours greater than 30 are paid at 2 times the hourly rate $((\text{Salary}/40 \text{ Hrs}) * \text{OT Hrs} * 2)$
 - v. Overtime is added to the base salary when calculating the total compensation (`calculateCompensation`)
- The Contract Employee Compensation is calculated with the following business rules within the `calculateCompensation` method:
 - i. Overtime is paid out at a rate of 1.5 times the hourly rate for hours greater than 40
 - 1. Example:
 - a. Rate of \$30/hr for the first 40 hours, \$45/hr for all hours beyond 40

Ensure the proper commenting for assignments is included at the top of your code. Post your code in pdf file format. Include all assignment on one pdf file.



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- Create a console application which instantiates a Fulltime employee and Contract Employee and:
 - i. Sets all properties
 - ii. Calls the saveEmployee and within the save executes the validation of Age, Rank and SSN
 - iii. A valid save state of each employee type will the display the details for the fulltime and contract employees by calling displayInformation
 - iv. Create one employee object that fails the validation checks for Age, Rank and SSN to test the validations

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Employee